

EXP. NO:
4 MATLAB SIMULATION OF SUPERHETERODYNE RECEIVER
FOR AM COMMUNICATION SYSTEM WITH FREQUENCY

MIXING AND FILTERING

Objectives

- i) To model and simulate the operation of a superheterodyne receiver for an AM communication system using MATLAB by implementing frequency mixing and intermediate frequency (IF) signal generation.
- ii) To recover the original message signal through intermediate frequency filtering and envelope detection, and analyze the waveforms of the message, carrier, AM, IF, and recovered signals.

CODE

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 10000;
fLO = 14550;

% Message signal
m = cos(2*pi*fm*t);

% Carrier signal
c = cos(2*pi*fc*t);

% AM signal (modulation index = 1)
am = (1 + m).*cos(2*pi*fc*t);

% Local oscillator
LO = cos(2*pi*fLO*t);

% Frequency mixing
mixed = am .* LO;

% IF filtering using FFT (no butter() required)
```

```

N = length(mixed);
f = (0:N-1)*(fs/N);

MIX_FFT = fft(mixed);

% Keep the IF band around 4550 Hz and its negative-frequency equivalent
fIF = abs(f);
fIF(f > fs/2) = fs - f(f > fs/2);

BW = 1500;
mask = abs(fIF - 4550) <= BW;

IF = real(ifft(MIX_FFT .* mask));

% Envelope detection using absolute value and moving average
env = abs(IF);
env = movmean(env, round(fs/2000));

% Remove DC component
env = env - mean(env);

% Plot results
figure('Name','Superheterodyne AM Receiver', ...
       'NumberTitle','off');

subplot(5,1,1);
plot(t,m);
title('Message Signal');
grid on;

subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
grid on;

subplot(5,1,3);
plot(t,am);
title('AM Signal');
grid on;

subplot(5,1,4);
plot(t,IF);
title('Intermediate Frequency (IF) Signal');
grid on;

```

```
subplot(5,1,5);  
plot(t,env);  
title('Recovered Signal');  
xlabel('Time (s)');  
grid on;
```

OUTPUT

